HTCS 6706 Data Analysis

Part 2: Final Report and Analysis

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Executive Summary

This report presents a comprehensive data analysis project focused on detecting and understanding phishing websites, utilising a dataset of 11,055 websites with 30 structural attributes. Three stages of analysis — descriptive, exploratory, and predictive — were applied to identify patterns, correlations, and predictive indicators that distinguish phishing websites from legitimate ones.

Descriptive Analysis revealed that phishing websites commonly exhibit specific characteristics such as abnormal SSL certificates, suspicious anchor links, long URLs, and frequent use of subdomains or hyphens.

Visualisation methods from **exploratory Analysis** confirmed that the attributes **SSLfinal_State** and **URL_of_Anchor** show the strongest correlation with phishing behaviour. These features consistently differentiated phishing from legitimate sites across various visualisations, including bar charts, boxplots, histograms, heatmaps, scatter plots, and pie charts.

Predictive Analysis applied three learning algorithms —Logistic Regression, Decision Tree, and Random Forest — to predict website legitimacy. All models achieved high accuracy (92–97%), with the **Random Forest model performing best** at **96.7% accuracy**, demonstrating robust classification performance and minimal misclassifications.

The Analysis concludes that **SSL validation** and **link structure** are the most critical indicators for phishing detection. Implementing automated monitoring systems that prioritise these attributes can enhance early phishing detection. The study demonstrates that data-driven predictive modelling is a practical approach for strengthening cybersecurity and mitigating web-based threats.

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Introduction

The purpose of this analysis report is to analyse phishing website behaviour using a data-driven approach and machine learning techniques with Jupyter Notebook. Phishing remains one of the most prevalent cybersecurity threats, exploiting users through fraudulent websites that mimic legitimate domains to steal credentials and sensitive data. This report will help readers identify and avoid phishing websites.

In the dataset, '**Training Dataset.arff**' (Mohammad, R., & McCluskey, 2015), it features 30 attributes to analyse 11,055 different websites, and there are three flags in each attribute: '-1': phishing, '0': suspicious, and '1': legitimate. The last 31st attribute, '**Result**', indicates whether a site is phishing (-1) or legitimate (1). In this report, I conducted **descriptive**, **exploratory**, and **predictive** analyses on this dataset to address the following cybersecurity questions:

1. What are the most common features of phishing websites in the dataset?
2. Are there any significant patterns or correlations among the feature set that help distinguish phishing sites from legitimate ones?
3. Can we build a machine learning model to predict phishing websites?

The descriptive analysis section provides a summary of the dataset, including both 'quantitative analysis' and 'qualitative analysis'. It quantifies and summarises attribute behaviour to identify the most common phishing characteristics.

The exploratory analysis section covers visualisation methods to uncover attribute patterns and correlations that distinguish phishing from legitimate sites.

The predictive analysis section covers three machine learning models that automatically predict phishing websites.

In the '**conclusion**' section, it summarises all the key findings from each Analysis and their impact on cybersecurity practices.

Descriptive Analysis

Descriptive Analysis is a data research technique to describe, demonstrate, or summarise a dataset (Villegas, n.d.). It can answer questions like “what is the average, median, quartiles, minimum, maximum, standard deviation, and frequency of phishing flags (-1) and legitimate flags (1) from both phishing and legitimate websites?”

This Analysis aims to help understand the general structure and characteristics of the dataset by performing various descriptive data analysis methods. It involves examining both **quantitative** and **qualitative** variables to identify common patterns, central tendencies, and distributions among website attributes.

The goal in this section is to answer the first cybersecurity question: ***“What are the most common features of phishing websites in the dataset?”*** Understanding these descriptive patterns will provide the foundation for deeper exploratory and predictive Analysis in later sections.

In the dataset ‘**Training Dataset.arff**’, there are three flags: ‘**-1**’ phishing, ‘**0**' suspicious, and '**1**’ legitimate. Out of 11,055 entries, the number of phishing websites is 4,898, and the number of legitimate websites is 6,157 from the ‘Result’ column. The dataset is moderately balanced, comprising 44.3% phishing and 55.7% legitimate websites, which enables the training of a reliable model.

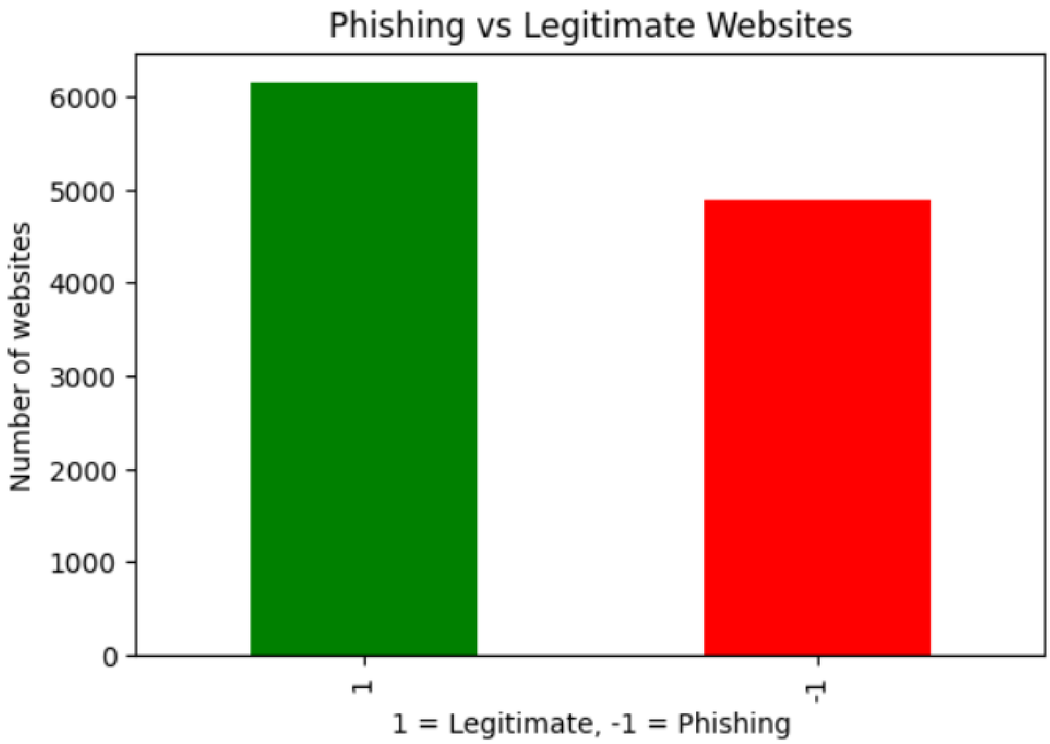


Figure 1. *Bar graph showing the number of phishing and legitimate websites*

	Phishing flags	Legitimate flags	Suspicious flags
having_IP_Address	3793	7262	0
URL_Length	8960	1960	135
Shortining_Service	1444	9611	0
having_At_Symbol	1655	9400	0
double_slash_redirecting	1429	9626	0
Prefix_Suffix	9590	1465	0
having_Sub_Domain	3363	4070	3622
SSLfinal_State	3557	6331	1167
Domain_registeration_length	7389	3666	0
Favicon	2053	9002	0
port	1502	9553	0
HTTPS_token	1796	9259	0
Request_URL	4495	6560	0
URL_of_Anchor	3282	2436	5337
Links_in_tags	3956	2650	4449
SFH	8440	1854	761
Submitting_to_email	2014	9041	0
Abnormal_URL	1629	9426	0
Redirect	0	1279	9776
on_mouseover	1315	9740	0
RightClick	476	10579	0
popUpWidnow	2137	8918	0
Iframe	1012	10043	0
age_of_domain	5189	5866	0
DNSRecord	3443	7612	0
web_traffic	2655	5831	2569
Page_Rank	8201	2854	0
Google_Index	1539	9516	0
Links_pointing_to_page	548	4351	6156
Statistical_report	1550	9505	0

Figure 2. *Frequency table of each flag in each attribute from all websites*

Quantitative Statistics

'**Quantitative'** or also known as **'Numerical variables**' are a type of variable in descriptive analysis that quantifies or measures the characteristics of an entity (Kumar, 2023). It describes **what the data looks like**. This section covers the Analysis of phishing websites with:

- Mean
- Median
- Mode

- Minimum, Maximum, and Range
- Quartiles and Interquartile Range
- Standard Deviation
- Correlation Coefficients

Mean

This analysis aims to determine the average number of phishing flags on each type of website (phishing or legitimate).

```
Mean for each flag in phishing websites are:
Phishing flags      1699.03
Legitimate flags    2667.87
Suspicious flags     531.10
dtype: float64

Mean for each flag in legitimate websites are:
Phishing flags      1581.37
Legitimate flags    3974.33
Suspicious flags     601.30
dtype: float64
```

Figure 3. *Mean value of each flag in each type of website*

Based on the results from Figure 3, the mean value of attributes flagged as phishing in phishing websites is **1699.03**, and in legitimate websites is **1581.37**. The mean values indicate that phishing websites generally contain more phishing flag features (mean = 1699) compared to legitimate websites (mean = 1581). However, phishing sites still exhibit a considerable number of legitimate flags (mean = 2668), showing that many fraudulent pages combine legitimate elements with deceptive characteristics to appear trustworthy. In contrast, legitimate websites maintain a much higher mean of legitimate flags (mean = 3974), reflecting consistent adherence to secure and authentic design patterns. The relatively similar counts of suspicious flags between both classes suggest that certain features remain ambiguous and may not distinctly separate phishing from legitimate behavior. Overall, these results highlight that phishing detection cannot rely on single attributes but must evaluate feature combinations and contextual patterns.

Median

This analysis aims to determine the middle number of phishing flags for each type of website.

```
Median for each flag in phishing websites are:
Phishing flags       939.0
Legitimate flags    3076.0
Suspicious flags       0.0
dtype: float64

Median for each flag in legitimate websites are:
Phishing flags      1053.0
Legitimate flags    4699.5
Suspicious flags       0.0
dtype: float64
```

Figure 4. *Median value of each flag in each type of website*

Based on the results from Figure 4, the median of attributes flagged as phishing in phishing websites is **939**, and in legitimate websites is **1053**. In phishing websites, the median number of phishing flags (939) is lower than the median number of legitimate flags (3076), indicating that while phishing pages display several deceptive characteristics, they also incorporate many legitimate features to appear trustworthy. In contrast, legitimate websites have a significantly higher median number of legitimate flags (4,699.5), indicating consistency. Interestingly, legitimate websites also show a small median number of phishing flags (1053), suggesting that occasional false positives may occur, where legitimate websites can exhibit certain phishing-like traits, such as long URLs or redirects. Both website categories display a median of zero suspicious flags, indicating that most attributes are distinctly classified as either phishing or legitimate. Overall, the median results reinforce that phishing detection must consider combinations of features rather than single indicators, as many malicious sites imitate legitimate characteristics.

Mode

This analysis aims to identify the flag that appears most frequently for each type of website. It will tell us what the **dominant behaviour** of phishing vs legitimate websites looks like.

Result	From Phishing Websites	From Legitimate Websites
SSLfinal_State	-1	1
Request_URL	-1	1
age_of_domain	-1	1
URL_of_Anchor	-1	0
Links_in_tags	-1	0
URL_Length	-1	-1
Prefix_Suffix	-1	-1
Domain_registeration_length	-1	-1
SFH	-1	-1
Page_Rank	-1	-1
having_Sub_Domain	0	1
web_traffic	0	1
Redirect	0	0
Links_pointing_to_page	0	0
having_IP_Address	1	1
Shortining_Service	1	1
having_At_Symbol	1	1
double_slash_redirecting	1	1
Favicon	1	1
port	1	1
HTTPS_token	1	1
Submitting_to_email	1	1
Abnormal_URL	1	1
on_mouseover	1	1
RightClick	1	1
popUpWidnow	1	1
Iframe	1	1
DNSRecord	1	1
Google_Index	1	1
Statistical_report	1	1

Figure 5. *Mode of every attribute in each type of website*

The attributes ‘**URL_Length**’, ‘**Prefix_Suffix**’, ‘**Domain_registration_length**’, ‘**SFH**’, and ‘**Page_Rank**’ are most often flagged as phishing in both categories. It means that some features that detect phishing can also appear for legitimate websites occasionally. These could be false positives or ambiguous indicators.

The attributes ‘**SSLfinal_State**’, ‘**Request_URL**’, and ‘**age_of_domain**’ are strongly phishing dominant as phishing flag (-1) is more frequent in phishing websites only, and legitimate flag (1) is more frequent in legitimate websites.

The rest of the attributes showed that legitimate flags are more frequent for both types of websites, indicating that they are not strong distinguishing features, as both websites exhibit the same typical safe behaviour most of the time.

Minimum, Maximum, and Range

This analysis aims to determine the minimum and maximum values of phishing flags for each type of website and then calculate the range between the maximum and minimum phishing flags.

```
Minimum for phishing flag in phishing websites is:
Phishing flags      0
Legitimate flags    0
Suspicious flags    0
dtype: int64

Maximum for phishing flag in phishing websites is:
Phishing flags      4898
Legitimate flags    4673
Suspicious flags    4296
dtype: int64

Minimum for phishing flag in legitimate websites is:
Phishing flags        0
Legitimate flags    677
Suspicious flags      0
dtype: int64

Maximum for phishing flag in legitimate websites is:
Phishing flags      4881
Legitimate flags    5906
Suspicious flags    5480
dtype: int64
```

Figure 6. *Minimum and Maximum number of each flag in phishing and legitimate websites*

```
Range for phishing flag in phishing websites is:
Phishing flags      4898
Legitimate flags    4673
Suspicious flags    4296
dtype: int64

Range for phishing flag in legitimate websites is:
Phishing flags      4881
Legitimate flags    5229
Suspicious flags    5480
dtype: int64
```

Figure 7. Range of each flag in phishing and legitimate websites

Figure 6 shows that all minimum values are 0. Some attributes are not represented in any specific category. For example, a few phishing flags may not appear on some legitimate websites and vice versa. This reinforces that **no single attribute** can classify a website as phishing; multiple indicators must be considered together.

In phishing websites, the maximum phishing flag count (4,898) and the maximum legitimate flag count (4,673) are both relatively high. It means that some phishing sites exhibit a strong presence of phishing indicators, while others heavily mimic legitimate features.

In legitimate websites, the maximum legitimate flag count (5906) is higher than the phishing flag count. Therefore, legitimate websites consistently display strong positive indicators with fewer phishing-like features.

Based on the results from Figure 7, the range of attributes flagged as phishing in phishing websites is **4898**, and in legitimate websites is **4881**. Some phishing websites show extreme phishing behaviour, while others are only partially deceptive. The extensive range of values confirms once again that phishing detection should rely on multiple features rather than isolated indicators, as both legitimate and malicious websites can vary widely in how they present their characteristics.

Quartiles and Interquartile Range

This analysis aims to determine the 1st and 3rd quartiles and then calculate the interquartile range between these two quartiles.

```
1st quartile for each flag in phishing websites are:
Phishing flags      671.50
Legitimate flags    860.75
Suspicious flags      0.00
Name: 0.25, dtype: float64

1st quartile for each flag in legitimate websites are:
Phishing flags       671.25
Legitimate flags    2358.25
Suspicious flags       0.00
Name: 0.25, dtype: float64

3rd quartile for each flag in phishing websites are:
Phishing flags      2664.25
Legitimate flags    4138.25
Suspicious flags     218.00
Name: 0.75, dtype: float64

3rd quartile for each flag in legitimate websites are:
Phishing flags      1796.25
Legitimate flags    5326.75
Suspicious flags      44.25
Name: 0.75, dtype: float64
```

Figure 8. *1st quartile and 3rd quartile of every attribute in each type of website*

```
Interquartile range for each flag in phishing websites are:
Phishing flags      1992.75
Legitimate flags    3277.50
Suspicious flags     218.00
dtype: float64

3rd quartile for each flag in legitimate websites are:
Phishing flags      1125.00
Legitimate flags    2968.50
Suspicious flags      44.25
dtype: float64
```

Figure 9. *Interquartile range of every attribute in each type of website*

Based on the results from Figure 9, the interquartile range of attributes flagged as phishing in phishing websites is **1992.75**, and in legitimate websites is **1125**. Therefore, the extensive range of phishing websites means they vary significantly in the number of phishing characteristics they display. Some are overtly malicious, while others appear semi-legitimate. Legitimate websites showed a smaller range, indicating that they consistently follow safe design practices, although a small subset may have attributes that superficially resemble those of phishing websites. Overall, the IQR results support the finding that phishing sites are more diverse and unpredictable in their design, while legitimate websites are structurally consistent and reliable.

Standard Deviation

This analysis aims to determine how spread out or dispersed the flags are from the mean value. Low STD means that most values are close to the average, making the data predictable and consistent. High STD means most values are spread far from the mean, showing high variability. So, attributes with low STD in both phishing and legitimate websites are likely to be linked to phishing websites.

```
Standard Deviation of Phishing Websites:
Phishing flags      1368.84
Legitimate flags    1625.27
Suspicious flags    1071.30
dtype: float64

Standard Deviation of Legitimate Websites:
Phishing flags      1466.03
Legitimate flags    1675.11
Suspicious flags    1370.55
dtype: float64
```

Figure 10. Standard Deviations from Phishing and Legitimate Websites

Based on the results from Figure 10, the STD of attributes flagged as phishing in phishing websites is **1368.84**, and in legitimate websites is **1466.03**. Therefore, all standard deviations are large compared to the minimums (often 0), indicating that the frequency of such flags across features is highly variable. Some features are flagged very frequently; others are hardly ever. This makes them harder to detect with simple rule-based systems, emphasising the need for machine learning models that can handle diverse feature combinations.

Correlation Coefficient

This Analysis is to find the strength of a linear relationship between:

- Phishing flags of attributes from phishing websites and phishing flags of attributes from legitimate websites.
- Phishing flags of attributes from phishing websites and legitimate flags of attributes from legitimate websites.

```
Correlation coefficient between phishing flags in phishing website and phishing flags in legitimate websites:
[[1.         0.79502542]
 [0.79502542 1.        ]]

Correlation coefficient between phishing flags in phishing website and legitimate flags in legitimate websites
[[ 1.         -0.62368204]
 [-0.62368204  1.        ]]
```

Figure 11. *Correlation coefficients*

Based on the results from Figure 11, it is revealed that:

- The correlation coefficient between phishing flags in phishing websites and phishing flags in legitimate websites is relatively close to +1 (0.79502542). This means that attributes with more phishing flags found on phishing websites tend to have more phishing flags on legitimate websites as well. That means even many legitimate websites' attributes are flagged as phishing, which may reflect feature overlap or false positives.
- The correlation coefficient between phishing flags in phishing websites and legitimate flags in legitimate websites is relatively close to -1 (0.62368204). This means that when phishing flags are common in phishing websites, legitimate flags are rare in legitimate websites, and vice versa.

```
Result                          1.000000
SSLfinal_State                  0.735814
URL_of_Anchor                   0.701207
web_traffic                     0.365118
Prefix_Suffix                   0.348606
having_Sub_Domain               0.304692
Request_URL                     0.253372
Links_in_tags                   0.250711
SFH                             0.219190
Google_Index                    0.128950
age_of_domain                   0.121496
Page_Rank                       0.104645
having_IP_Address               0.094160
Statistical_report              0.079857
DNSRecord                       0.075718
URL_Length                      0.053332
having_At_Symbol                0.052948
on_mouseover                    0.041838
Links_pointing_to_page          0.041085
port                            0.036419
Submitting_to_email             0.018249
RightClick                      0.012653
popUpWidnow                     0.000086
Favicon                        -0.000280
Iframe                         -0.003394
Redirect                       -0.020113
double_slash_redirecting       -0.038608
HTTPS_token                    -0.039854
Abnormal_URL                   -0.060488
Shortining_Service             -0.067966
Domain_registeration_length    -0.225789
Name: Result, dtype: float64
```

Figure 12. *Spearman Correlations of each attribute*

The Spearman correlation coefficient measures the strength and direction of monotonic (rank-based) association between variables. Values range:

- 1.0: Perfect positive relationship.
- −1.0: Perfect negative relationship.
- 0: No monotonic relationship.

Results from Figure 12 show that the attributes ‘**SSLfinal_State (0.74)**’ and ‘**URL_of_Anchor (0.70)**’ have a significantly high positive correlation with the ‘Result’ attribute; therefore, these two attributes are the strongest candidates for key indicators in identifying phishing websites. Moderate correlations were also found in ‘**web_traffic**’, ‘**Prefix_Suffix**’, and ‘**having_Sub_Domain**’, suggesting that legitimate sites generally have high traffic, few subdomains, and no hyphens in their domain names.

Qualitative Statistics

‘**Qualitative**’ or also known as ‘**Categorical variables**’ are variables in descriptive Analysis that can be grouped into categories, based on specific characteristics. They are used in statistical Analysis to measure the relationships between different factors in a study (Kumar, 2023). It explains **what those numbers mean** in the context of cybersecurity**.** This section covers the Analysis of phishing websites with:

- Frequency tables
- Percentage

Frequency tables

The purpose of this Analysis is to find out the most common attributes that phishing websites have and the least common attributes that legitimate websites have. The more phishing flags appear from phishing websites and the fewer from legitimate websites, the closer the relationship between the attributes and phishing websites.

Number of each flags from phishing websites:

	Phishing flags	Legitimate flags
Prefix_Suffix	4898	0
SFH	4238	397
URL_Length	4079	736
Page_Rank	3885	1013
URL_of_Anchor	3246	150
SSLfinal_State	3051	701
Domain_registeration_length	2690	2208
Request_URL	2675	2223
age_of_domain	2632	2266
Links_in_tags	2387	767
having_IP_Address	1926	2972
having_Sub_Domain	1836	810
DNSRecord	1718	3180
web_traffic	1673	1507
popUpWidnow	947	3951
Submitting_to_email	931	3967
Google_Index	927	3971
Favicon	909	3989
Statistical_report	839	4059
having_At_Symbol	837	4061
port	734	4164
HTTPS_token	715	4183
on_mouseover	657	4241
Abnormal_URL	604	4294
double_slash_redirecting	562	4336
Shortining_Service	514	4384
Iframe	443	4455
RightClick	225	4673
Links_pointing_to_page	193	1776
Redirect	0	602

Number of each flags from legitimate websites:

	Phishing flags	Legitimate flags
Redirect	0	677
URL_of_Anchor	36	2286
RightClick	251	5906
Links_pointing_to_page	355	2575
SSLfinal_State	506	5630
Iframe	569	5588
Google_Index	612	5545
on_mouseover	658	5499
Statistical_report	711	5446
port	768	5389
having_At_Symbol	818	5339
double_slash_redirecting	867	5290
Shortining_Service	930	5227
web_traffic	982	4324
Abnormal_URL	1025	5132
HTTPS_token	1081	5076
Submitting_to_email	1083	5074
Favicon	1144	5013
popUpWidnow	1190	4967
having_Sub_Domain	1527	3260
Links_in_tags	1569	1883
DNSRecord	1725	4432
Request_URL	1820	4337
having_IP_Address	1867	4290
age_of_domain	2557	3600
SFH	4202	1457
Page_Rank	4316	1841
Prefix_Suffix	4692	1465
Domain_registeration_length	4699	1458
URL_Length	4881	1224

Figure 13. *Frequencies of each flag in each attribute from phishing and legitimate websites*

Frequency analysis showed us that:

The attributes ‘**Prefix_Suffix**’, ‘**SFH**’, ‘**URL_Length**’, ‘**Page_Rank**’, ‘**SSLfinal_State**', and '**URL_of_Anchor**’'s phishing flags appeared significantly high in phishing websites compared to legitimate flags.

The attributes ‘**Redirect**’, ‘**URL_of_Anchor**’, ‘**RightClick**’, ‘**Link_poiting_to_page**’, and ‘**SSLfinal_State**’'s phishing flags appeared significantly lower in legitimate websites compared to phishing flags.

Although attributes ‘**Prefix_Suffix**’, ‘**SFH**’, **URL_Length**’, and ‘**Page_Rank**’ phishing flags appear frequently in phishing websites, they also appear frequently on legitimate websites. Hence, these four attributes are weak in detecting phishing websites.

<u>Percentage</u>

The purpose of this Analysis is to express the frequency table from above as a percentage.

Percentage of each flags from phishing websites:

	Phishing flags	Legitimate flags
Prefix_Suffix	100.00	0.00
SFH	86.53	8.11
URL_Length	83.28	15.03
Page_Rank	79.32	20.68
URL_of_Anchor	66.27	3.06
SSLfinal_State	62.29	14.31
Domain_registeration_length	54.92	45.08
Request_URL	54.61	45.39
age_of_domain	53.74	46.26
Links_in_tags	48.73	15.66
having_IP_Address	39.32	60.68
having_Sub_Domain	37.48	16.54
DNSRecord	35.08	64.92
web_traffic	34.16	30.77
popUpWidnow	19.33	80.67
Submitting_to_email	19.01	80.99
Google_Index	18.93	81.07
Favicon	18.56	81.44
Statistical_report	17.13	82.87
having_At_Symbol	17.09	82.91
port	14.99	85.01
HTTPS_token	14.60	85.40
on_mouseover	13.41	86.59
Abnormal_URL	12.33	87.67
double_slash_redirecting	11.47	88.53
Shortining_Service	10.49	89.51
Iframe	9.04	90.96
RightClick	4.59	95.41
Links_pointing_to_page	3.94	36.26
Redirect	0.00	12.29

Percentage of each flags from legitimate websites:

	Phishing flags	Legitimate flags
Redirect	0.00	11.00
URL_of_Anchor	0.58	37.13
RightClick	4.08	95.92
Links_pointing_to_page	5.77	41.82
SSLfinal_State	8.22	91.44
Iframe	9.24	90.76
Google_Index	9.94	90.06
on_mouseover	10.69	89.31
Statistical_report	11.55	88.45
port	12.47	87.53
having_At_Symbol	13.29	86.71
double_slash_redirecting	14.08	85.92
Shortining_Service	15.10	84.90
web_traffic	15.95	70.23
Abnormal_URL	16.65	83.35
HTTPS_token	17.56	82.44
Submitting_to_email	17.59	82.41
Favicon	18.58	81.42
popUpWidnow	19.33	80.67
having_Sub_Domain	24.80	52.95
Links_in_tags	25.48	30.58
DNSRecord	28.02	71.98
Request_URL	29.56	70.44
having_IP_Address	30.32	69.68
age_of_domain	41.53	58.47
SFH	68.25	23.66
Page_Rank	70.10	29.90
Prefix_Suffix	76.21	23.79
Domain_registeration_length	76.32	23.68
URL_Length	79.28	19.88

Figure 14. *Percentages of each flag in each attribute from phishing and legitimate websites*

From the phishing websites, the top five attributes that had the most phishing flags from Figure 14 are:

1. Prefix_Suffix (4898 with 100%)
2. SFH (4238 with 86.53%)
3. URL_Length (4079 with 83.28%)
4. Page_Rank (3885 with 79.32%)
5. URL_of_Anchor (3246 with 66.27%)

From the legitimate websites, the top five attributes that had the fewest phishing flags from Figure 14 are:

1. Redirect (0 with 0%)
2. URL_of_Anchor (36 with 0.58%)
3. RightClick (251 with 4.08%)
4. Links_pointing_to_page (355 with 5.77%)
5. SSLfinal_State (506 with 8.22%)

So, the attributes that showed both high phishing flags frequency in phishing websites and low phishing flags frequency in legitimate websites are:

- URL_of_Anchor
- SSLfinal_State

The overall results from quantitative and qualitative Analysis indicate that attributes '**SSLfinal_State**' and '**URL_of_Anchor**' are the strongest indicators in identifying phishing websites, suggesting that valid SSL certificates and proper internal linking structures are strong indicators of legitimate websites.

Exploratory Analysis

Exploratory Analysis is another data research technique to analyse and investigate data sets and summarise their main characteristics, often with data visualisation methods ("What is exploratory data analysis (EDA)?", n.d.).

The aim is to **identify the patterns or relationships present in the data**. This involves using visual methods such as 'Boxplots', 'Histograms', 'Density Functions', 'Scatter Plots', and 'Pie Charts'.

The goal here is to answer the second cybersecurity question: ***"Are there any significant patterns or correlations among the feature set that help distinguish phishing sites from legitimate ones?"*** with various exploratory analysis methods. This section covers visual Analysis of phishing websites with:

- **Bar Chart**: For frequency comparison
- **Boxplots**: For spread and variability
- **Histograms**: For distribution patterns
- **Heatmap**: For correlation between features
- **Scatter Plot**: For the relationship between key attributes
- **Pie Charts**: Overall Distribution

Bar Charts

Bar Charts show which attributes are flagged as phishing most frequently in phishing and legitimate websites.

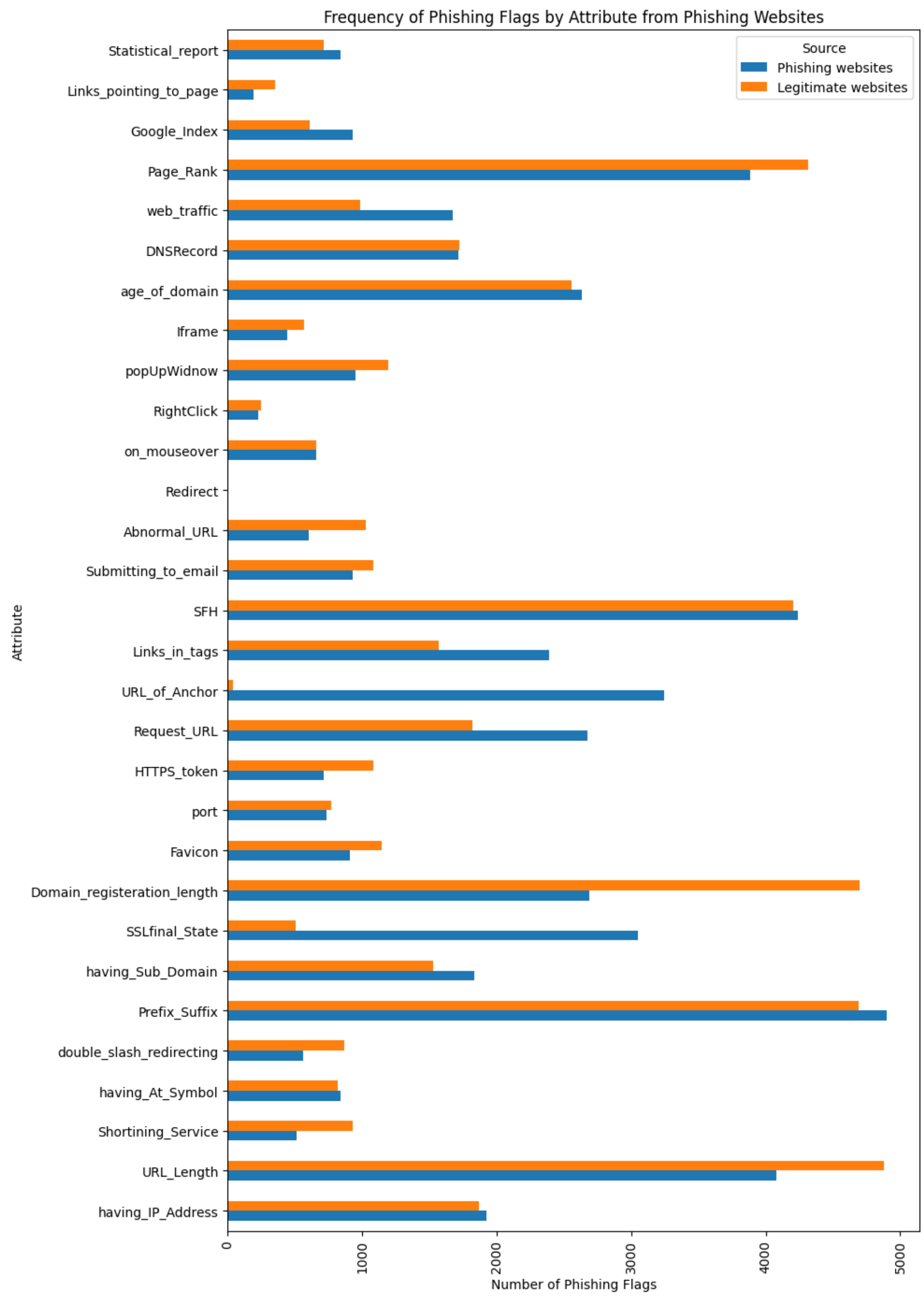


Figure 15. *Bar graph* showing f*requencies of phishing flags in each attribute from phishing and legitimate websites*

The bar chart from Figure 15 reveals that features such as '**Prefix_Suffix**', '**URL_Length**', '**Page_Rank**', and '**SFH**' occur most frequently as phishing indicators in phishing websites, which could mean they are the strongest candidates for identifying phishing websites. In contrast, attributes '**URL_Length**', '**Domain_registration_length**', and '**Prefix_Suffix**' occur more frequently as phishing indicators on legitimate websites than on phishing websites, which could result in the strongest false positives.

Some features, like '**having_IP_Address**', '**Shortining_Service**', or '**Abnormal_URL**', show similar frequencies for both website types. This means these attributes are less helpful in distinguishing phishing from legitimate sites based solely on the phishing flag.

The largest blue bars (phishing websites) are compared to the smaller or nearly absent orange bars (legitimate websites), which identify attributes most unique to phishing.

Boxplots

Boxplots are used to analyse the distribution of phishing flags with quantitative variables, 'minimum', '1st quartile', 'median', '3rd quartile', and 'maximum' in each type of website.

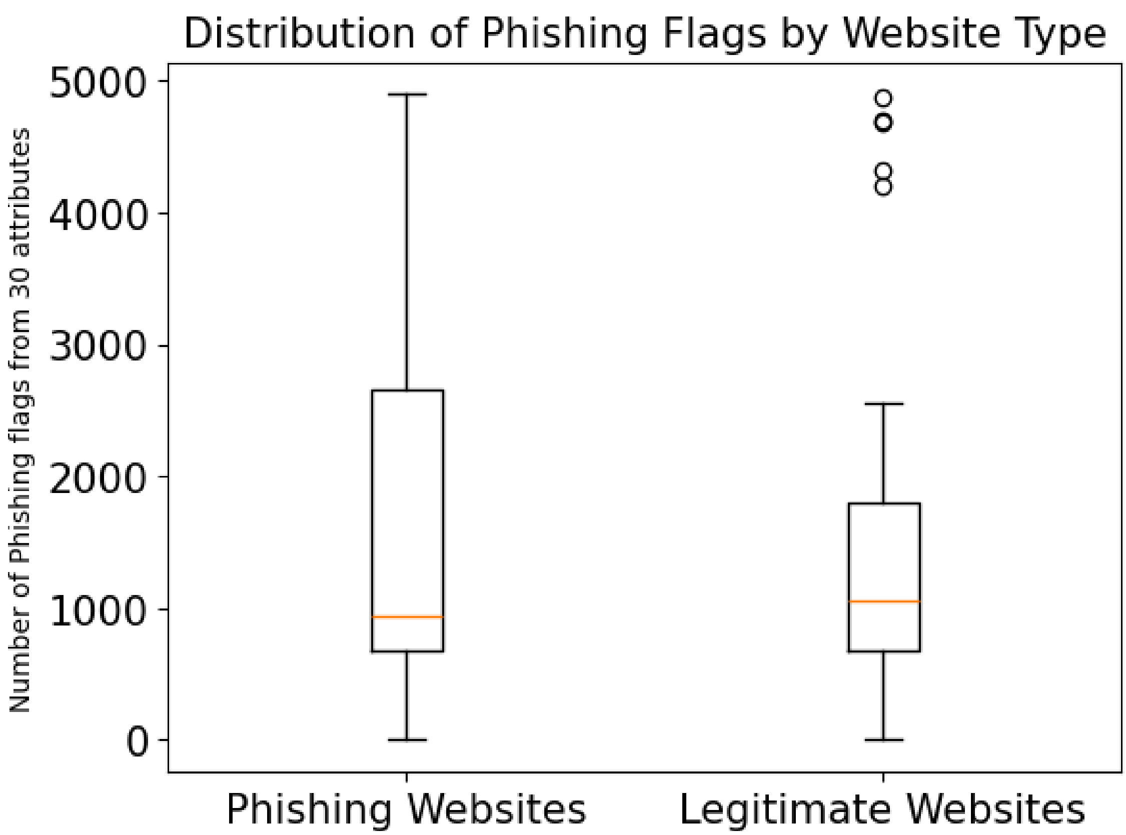


Figure 16. *Boxplots showing quantitative variables of phishing flags from phishing and legitimate websites*

The boxplots in Figure 16 show that, in phishing websites, the median (represented by the orange line) of phishing flags is approximately 1,000. This means that most phishing sites have moderate to high numbers of phishing-indicating features; some have very high numbers (close to 5,000), while others have low numbers.

On legitimate websites, the range is smaller compared to phishing websites, and the box is more compressed; the median is also around 1,000, but the upper bound is significantly lower than that of phishing websites. A few outliers have high numbers, but most legitimate sites have fewer phishing flags.

Histograms

Histograms are used to show how the flags of the top two key indicators from descriptive Analysis, '**SSLfinal_State**' and '**URL_of_Anchor**', are distributed differently between website types.

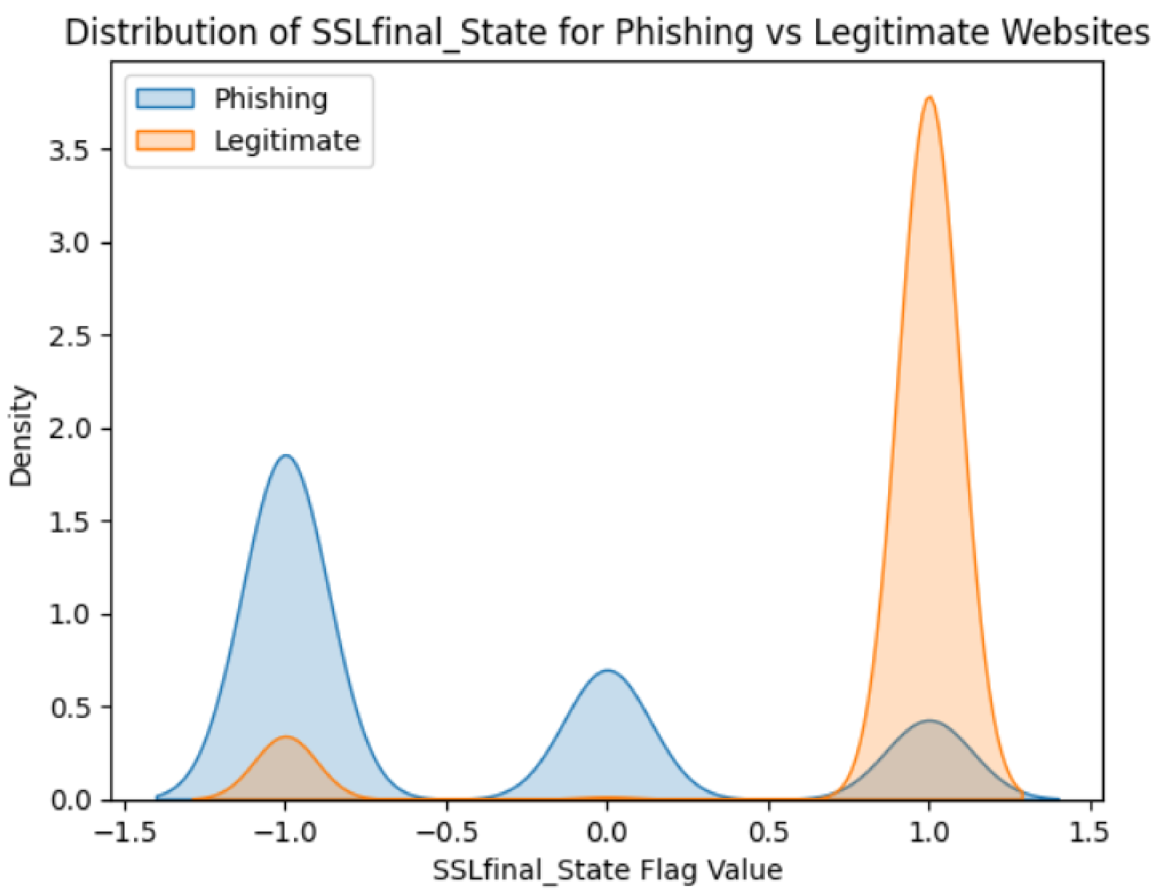


Figure 17. *Histogram / Density function showing flag distribution of SSLfinal_State*

There is a substantial spike at -1 for phishing websites (blue), while legitimate websites (orange) show very little density there. This indicates that most phishing websites have an **SSLfinal_State** flag value of -1, meaning they typically have an untrustworthy, missing, or abnormal SSL certificate.

Conversely, legitimate websites exhibit a distinct peak at +1, whereas phishing websites have very little density at this value. This means most legitimate websites have a secure and trusted SSL certificate.

This confirms that **SSLfinal_State** is a strong indicator of phishing.

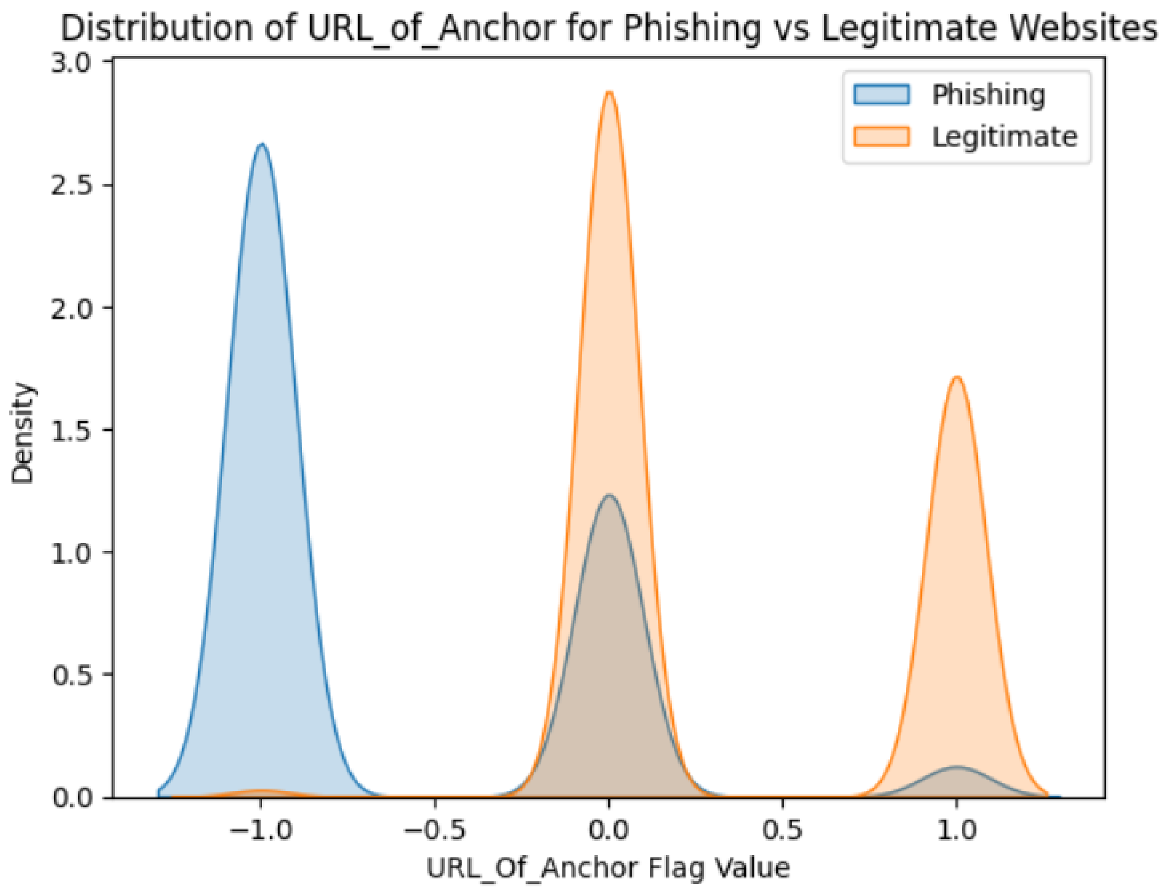


Figure 18. *Histogram / Density function showing flag distribution of URL_of_Anchor*

The density function in Figure 18 suggests that most phishing websites have a **URL_of_Anchor** flag value of -1, indicating a suspicious or abnormal anchor tag (e.g., many anchor links on the site point to potentially dangerous or misleading destinations).

In legitimate websites, there is also a clear peak at +1, suggesting a notable proportion of legitimate sites have trustworthy or normal values for this attribute. The density at -1 is significantly lower, indicating that it is rare for legitimate sites to have suspicious anchor tags.

This confirms that **URL_of_Anchor** is also a strong indicator of phishing.

Heatmap

A heatmap is used to visualise relationships between multiple attributes. Red dots indicate a strong positive correlation, meaning that both attributes increase/decrease together. Blue dots indicate a strong negative correlation, where one attribute increases and the other tends to decrease, indicating that the two attributes are weakly or not related at all.

To identify which significant patterns or correlations among the dataset distinguish phishing sites from legitimate ones, we look for attributes with a strong positive correlation with 'Result'.

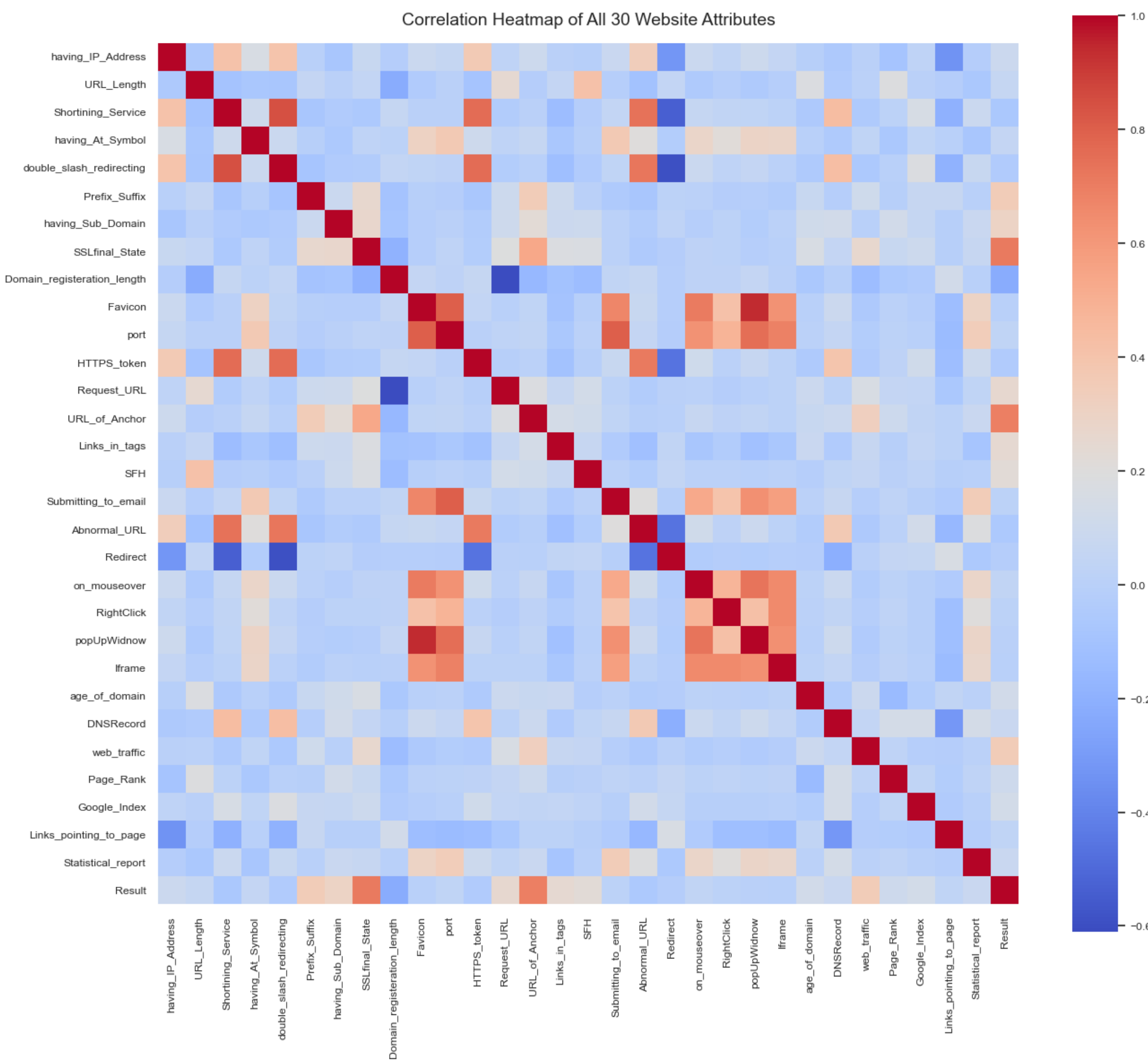


Figure 19. *Heatmap showing the relationship between 30 attributes and the Result (Type of website)*

From the figure 19 heatmap, there are two strong orange dots (strong positive relation) with 'Result'. It suggests that once again, the attributes, '**SSLfinal_State**' and '**URL_of_Anchor**' are strongly related to the type of website – more legitimate flags in legitimate websites and more phishing flags in phishing websites.

'**Domain_registration_length**' showed the strongest blue dot, indicating the weakest relationship with the Result.

The remaining attributes in relation to 'Result' are light-colored, indicating low correlation.

Scatter Plot

Scatter Plots are used to find any relationship between two dimensions (bivariate) in data. The red line is the line of equality, meaning attributes that sit on the line have equal counts of phishing flags in both types of sites. We want to identify attributes that are spread far away from this line.

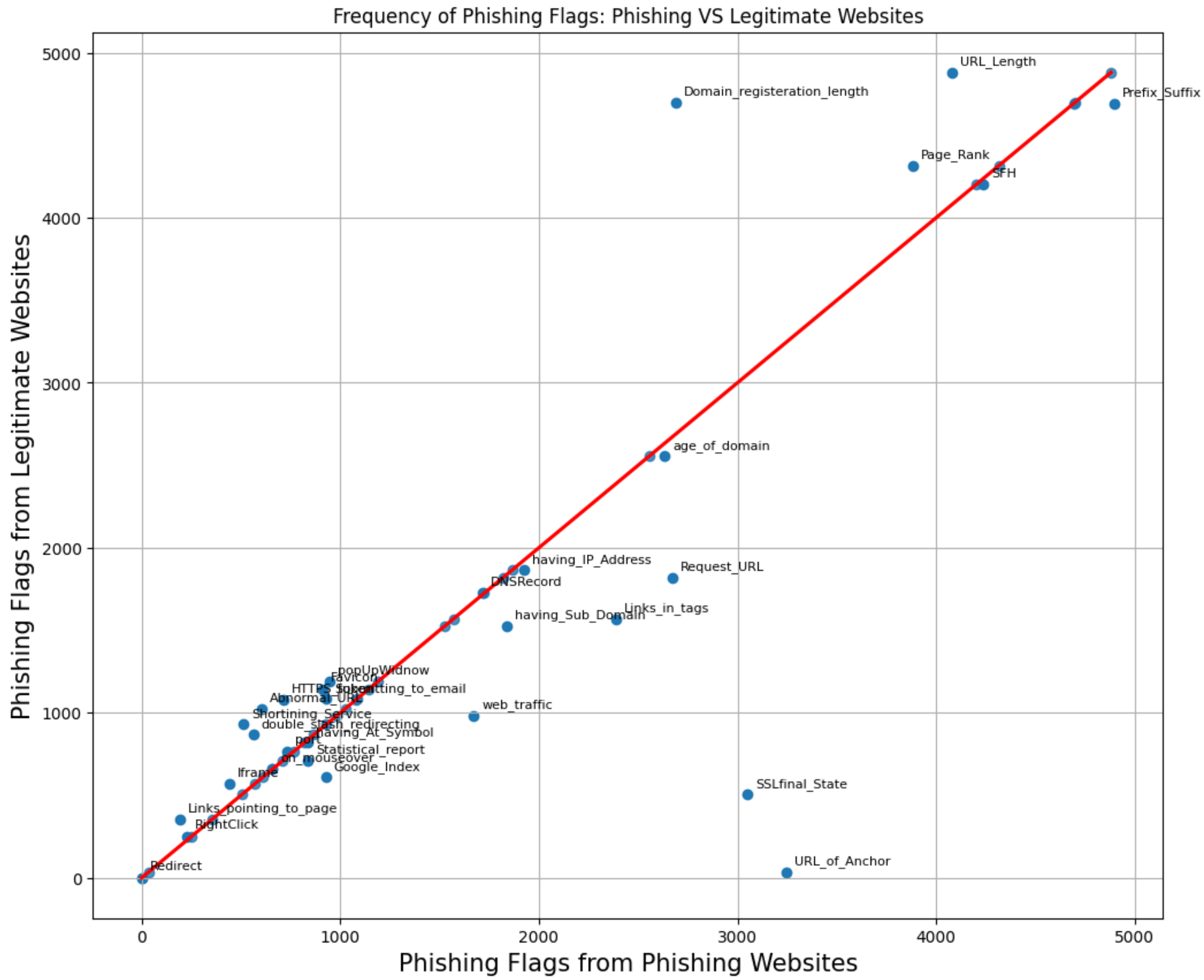


Figure 20. *A scatter plot showing the relationship between phishing flags in phishing and legitimate websites*

Figure 20 scatter plots suggest that the attributes ‘**Domain_registration_length’**, ‘**URL_Length**', and '**Page_Rank**’ are located far away to the top from the line of equality, meaning that they are highly flagged as false positives.

‘**SSLfinal_State**’ and ‘**URL_of_Anchor**’ are located far away to the bottom from the line of equality, meaning that they are highly flagged as a strong indicator for phishing websites.

The remaining attributes are located either closer to or precisely on the line of equality and do not provide helpful information for identifying patterns of phishing websites.

Pie Charts

The pie chart shows the proportions of each flag in phishing and legitimate websites. In this Analysis, I have examined two of the strongest phishing indicators, '**SSLfinal_State**' and '**URL_of_Anchor**', to see how these flags contribute to the overall analysis.

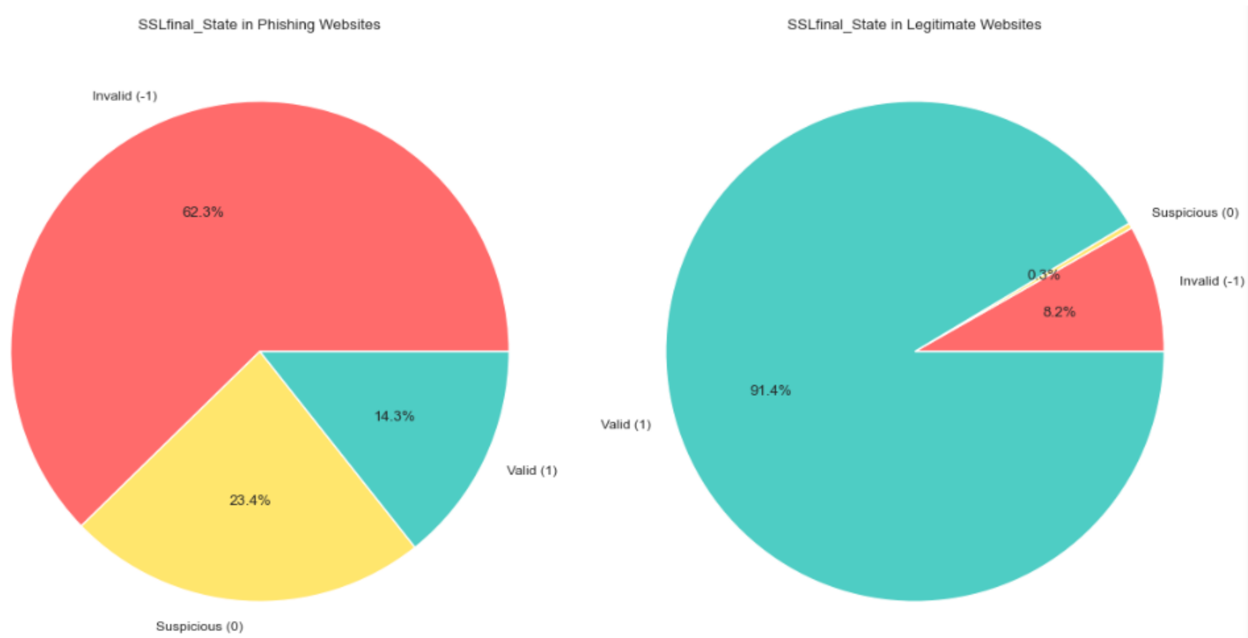


Figure 21. *Pie chart showing the frequency of each flag of SSLfinal_State in each type of website*

Figure 21 shows that phishing sites have mostly invalid SSL (62.3%); legitimate sites have mostly valid SSL (91.4%). This suggests that '**SSLfinal_State**' is a powerful distinguishing feature,

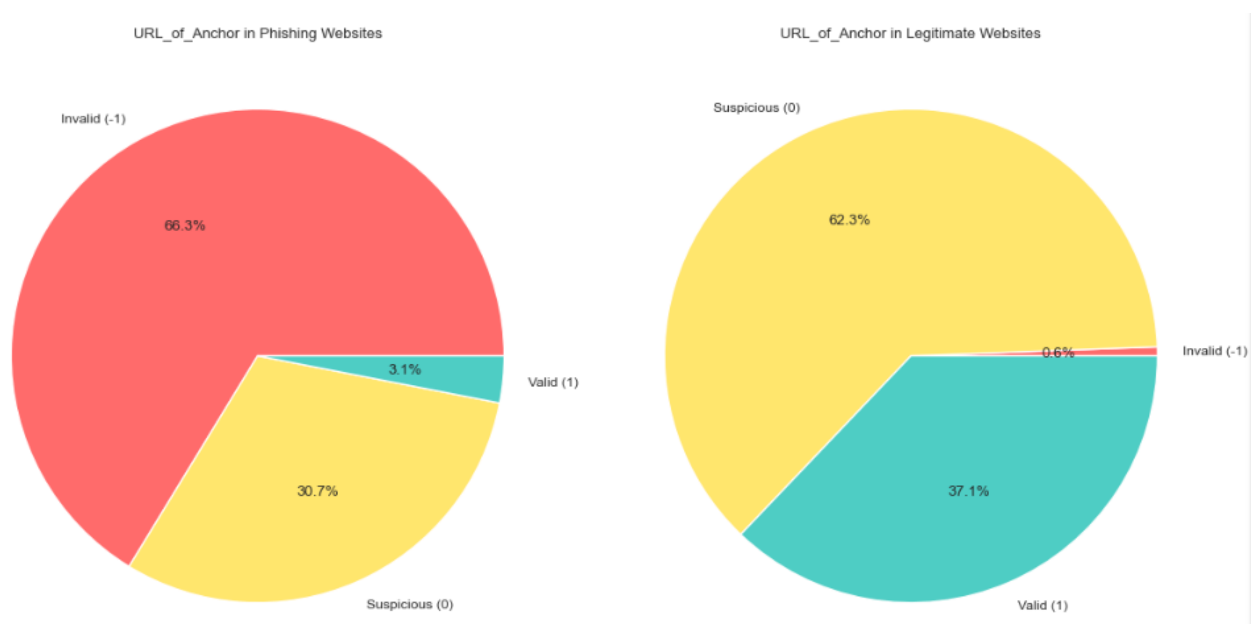


Figure 22. *Pie chart showing the frequency of each flag of URL_of_Anchor in each type of website*

Figure 22 shows that phishing websites have an invalid anchor flag, meaning most of their anchor links are suspicious (66.3%), and almost none (0.6%) of legitimate sites have invalid anchor flags. This suggests that '**URL_of_Anchor**' is also a powerful distinguishing feature,

The patterns from the six exploratory analyses show that bar charts, boxplots, histograms (and density plots), heatmaps, scatter plots, and pie charts consistently show that specific website attributes are powerful indicators for distinguishing phishing from legitimate websites.

The exploratory Analysis demonstrates that a small group of website attributes, especially SSL and anchor link validation, are highly effective for distinguishing phishing from legitimate sites. Visualisation tools reveal that most phishing websites are flagged as suspicious or untrusted by these features, while most legitimate sites are not. Relationships between features are mostly weak, supporting the use of multiple independent indicators in detection models. Among all visual methods, **histograms** and **heatmaps** most clearly revealed the strong influence of SSLfinal_State and URL_of_Anchor on phishing behaviour.

Predictive Analysis

Predictive Analysis is the practice of examining data to answer questions, identify trends, and extract insights. It predicts what might happen in the future (Cote, 2021).

The goal of this section is to address the third cybersecurity question: "***Can we build a machine learning model to predict phishing websites?***" using various predictive analysis methods. This section covers exploring several predictive analysis models on patterns identified in the exploratory data analysis*:*

- **Logistic Regression**: A statistical model for binary classification.
- **Decision Tree**: A Machine learning algorithm used for classification and prediction.
- **Random Forest**: Another machine learning algorithm that combines multiple decision trees to make more accurate and stable predictions.

Logistic Regression

Logistic Regression is a statistical method used to model the probability of a binary outcome. Outcome variables can take only two values, such as yes/no, true/false. In this Analysis, it takes Phishing (–1) or Legitimate (1). It works by identifying a mathematical relationship between **independent variables** (attributes such as SSLfinal_State, URL_Length, Prefix_Suffix, etc.) and a **dependent variable, '**Result'.

```
Accuracy: 0.924468566259611

Classification Report:
              precision    recall  f1-score   support

          -1       0.92      0.90      0.91       956
           1       0.93      0.94      0.93      1255

    accuracy                           0.92      2211
   macro avg       0.92      0.92      0.92      2211
weighted avg       0.92      0.92      0.92      2211


Confusion Matrix:
 [[ 865   91]
 [  76 1179]]

Feature Importance (Coefficients):
 URL_of_Anchor             3.197916
Prefix_Suffix             2.865164
SSLfinal_State            1.606730
Links_pointing_to_page    0.894304
SFH                       0.859545
Links_in_tags             0.843157
web_traffic               0.806469
Google_Index              0.679236
having_IP_Address         0.677274
having_Sub_Domain         0.615319
dtype: float64
```

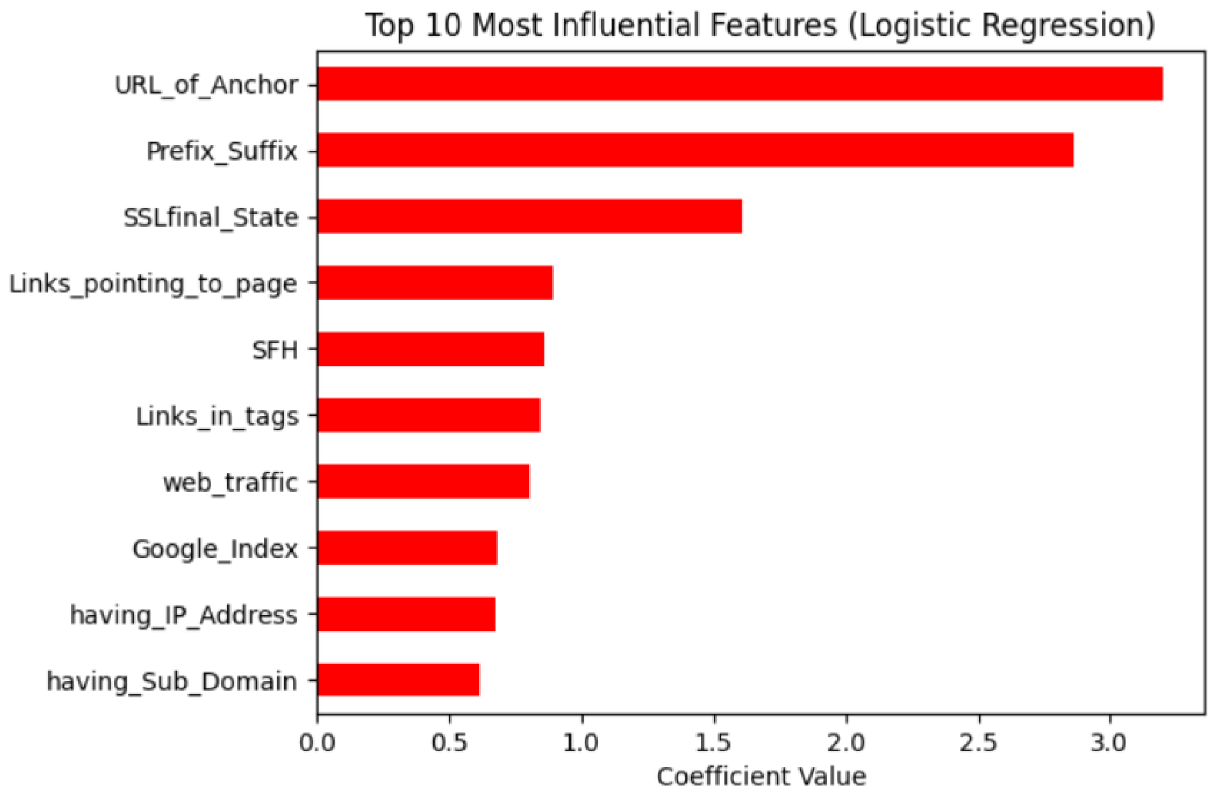


Figure 23. Logistic Regression Evaluation

The **Logistic Regression** resulted in an overall accuracy of 92.4%, successfully distinguishing between phishing websites and legitimate ones.

- **Precision** (Phishing: 0.92, Legitimate: 0.93):
- 92% of the sites predicted as phishing were truly phishing.
- **Recall** (Phishing: 0.90, Legitimate: 0.94):
- The model caught 90% of all phishing sites.
- **F1-score** (Phishing: 0.91, Legitimate: 0.93):
- The balance of precision and recall is powerful for both types, so the model shows reliable overall detection
- **Confusion Matrix**:
    - True phishing detected: 865 (correct)
    - Phishing misclassified as legitimate: 91
    - True legitimate detected: 1179 (correct)

- Legitimate misclassified as phishing: 76

This means most predictions are correct, with only a small number of misclassifications.

- **Feature Importance (Coefficients) & Bar chart**:

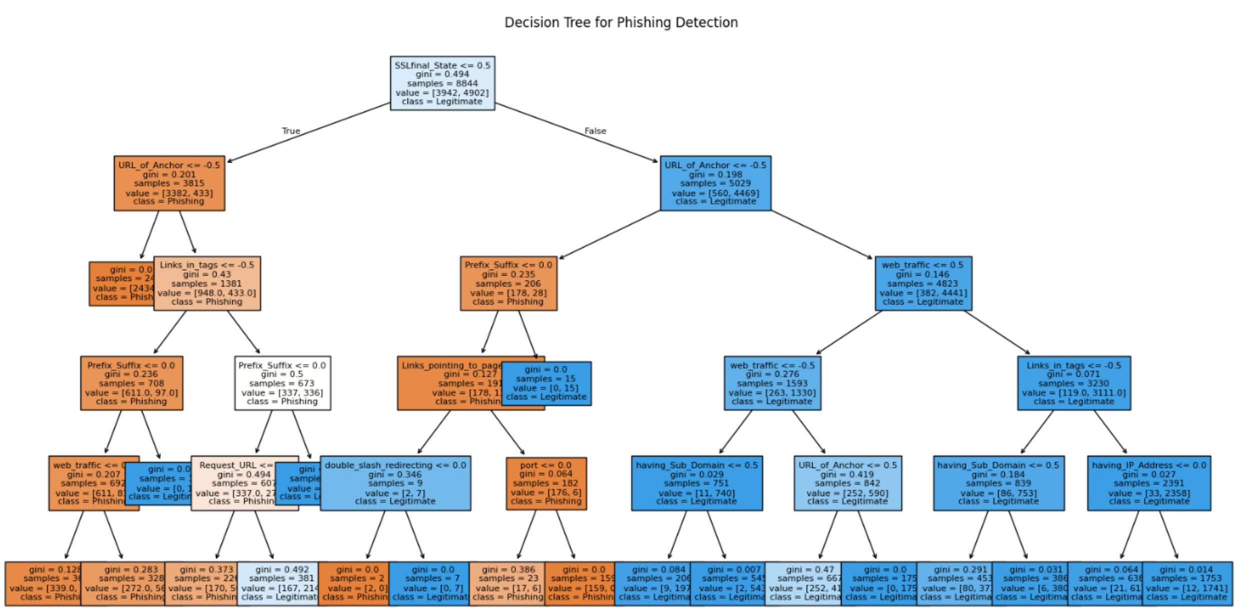
  Top five predictors:

  - URL_of_Anchor (most influential)
  - Prefix_Suffix
  - SSLfinal_State
  - Links_pointing_to_page
  - SFH
  - Features with higher coefficients contribute more to predicting phishing. In this model, **'URL_of_Anchor'** and '**Prefix_Suffix'** are far above the rest, meaning they are the strongest predictors of website legitimacy.

Decision Tree

A decision tree is a type of supervised machine learning algorithm used for both classification and regression tasks. It works by repeatedly splitting data into branches based on the value of input features, forming a tree-like structure that leads to decisions or predictions at the final leaf nodes.

```
Accuracy: 0.9249208502939846

Classification Report:
              precision    recall  f1-score   support

          -1       0.96      0.86      0.91       956
           1       0.90      0.98      0.94      1255

    accuracy                           0.92      2211
   macro avg       0.93      0.92      0.92      2211
weighted avg       0.93      0.92      0.92      2211


Confusion Matrix:
 [[ 820  136]
 [  30 1225]]

Feature Importance (Coefficients):
 SSLfinal_State              0.765174
URL_of_Anchor               0.133909
web_traffic                 0.030236
Links_in_tags               0.029448
Prefix_Suffix               0.025019
Request_URL                 0.008205
having_Sub_Domain           0.003314
Links_pointing_to_page      0.002793
double_slash_redirecting    0.000913
port                        0.000803
dtype: float64
```



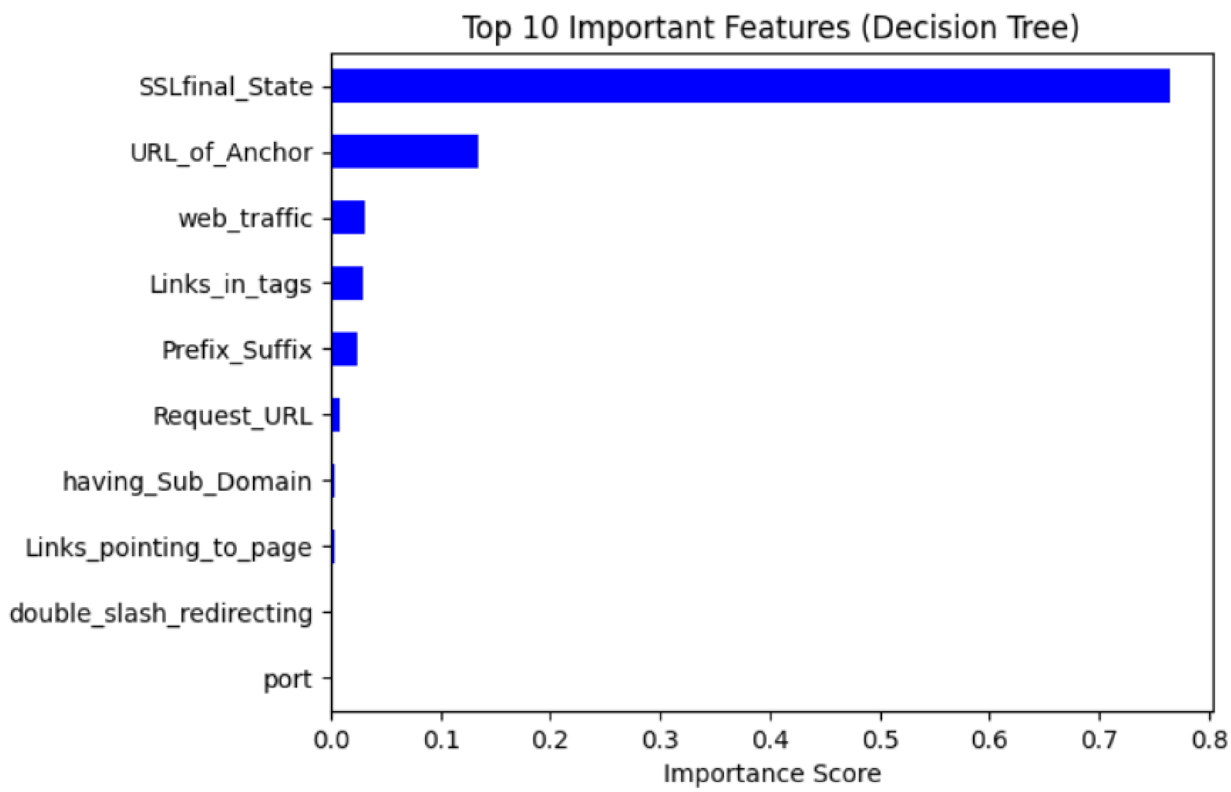


Figure 24. Decision Tree Evaluation

The **Decision Tree** resulted in an overall accuracy of 92.5%—slightly more accurate than Logistic Regression, but only by a tiny margin.

- **Precision** (Phishing: 0.96, Legitimate:0.90)**:**
  Fewer false positives and more phishing detection than Logistic Regression.
- **Recall** (Phishing: 0.86, Legitimate:0.98)**:**
  The model captures 86% of all actual phishing sites, but performs better than Logistic Regression in capturing legitimate sites, which means phishing site sensitivity is lower than that of Logistic Regression.
- **F1-score** (Phishing: 0.91, Legitimate:0.94)**:**
  Good balance as Logistic Regression.
- **Confusion Matrix:**
  - True phishing detected: 820 (correct)
  - Phishing misclassified as legitimate: 136
  - True legitimate detected: 1225 (correct)
  - Legitimate misclassified as phishing: 30
- **Feature Importance (Coefficients) & Bar chart:**
  Top five predictors:
  - SSlfinal_State (most influential)
  - URL_of_Anchor
  - web_traffic

  - Links_in_tags
  - Prefix_Suffix
- **Decision Tree:**
  The tree starts by splitting on the attribute SSLfinal_State (Is SSLfinal_State ≤ 0.5?). This means the model identified SSL status as the most critical initial indicator for phishing classification, followed by URL_of_Anchor.
  If a website fails a key check (e.g., has a suspicious SSL state, invalid anchor, etc.), it is classified as "phishing". If it passes the main checks, it is more likely to be classified as legitimate.

Random Forest

Random Forest is an ensemble learning algorithm that builds multiple Decision Trees and combines their results, making more accurate and stable predictions.

```
Accuracy: 0.9669832654907282

Confusion Matrix:
 [[ 909   47]
 [  26 1229]]

Classification Report:
               precision    recall  f1-score   support

          -1       0.97      0.95      0.96       956
           1       0.96      0.98      0.97      1255

    accuracy                           0.97      2211
   macro avg       0.97      0.97      0.97      2211
weighted avg       0.97      0.97      0.97      2211


Feature Importance (Coefficients):
 SSLfinal_State                0.326228
URL_of_Anchor                 0.245541
web_traffic                   0.069664
having_Sub_Domain             0.064523
Links_in_tags                 0.043273
Prefix_Suffix                 0.037831
SFH                           0.020350
Request_URL                   0.019330
Links_pointing_to_page        0.018834
Domain_registeration_length   0.017556
dtype: float64
```

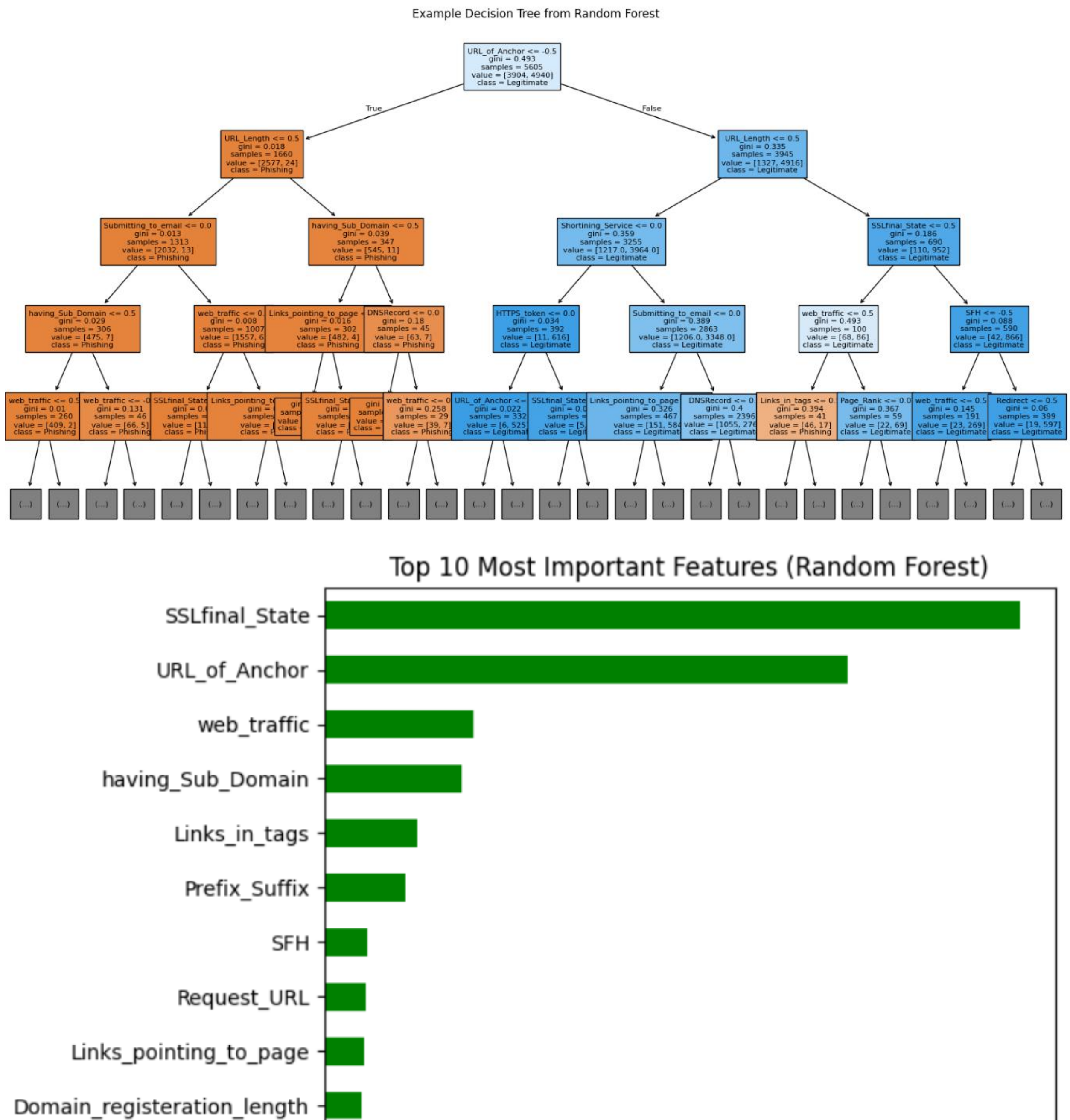


Figure 25. Random Forest Evaluation

The **Random Forest** yielded an overall accuracy of 96.7%, outperforming both Logistic Regression and a single decision tree.

- **Precision** (Phishing: 0.97, Legitimate:0.96):
  This model accurately predicts 97% of phishing websites.

- **Recall** (Phishing: 0.95, Legitimate:0.98):
  It detects 95% of all true phishing sites.
- **F1-score** (Phishing: 0.96, Legitimate:0.97):
  Stronger overall, balanced performance than the other two models.
- **Confusion Matrix**:
  - True phishing detected: 909 (correct)
  - Phishing misclassified as legitimate: 47
  - True legitimate detected: 1229 (correct)
  - Legitimate misclassified as phishing: 26
- **Feature Importance (Coefficients) & Bar chart**:
  Top five predictors:
  - SSlfinal_State (most influential)
  - URL_of_Anchor
  - web_traffic
  - having_Sub_Domain
  - Links_in_tags
- **Random Forest**:
  One of the visualised Decision Tree examples above demonstrates how the Random Forest model classifies websites based on feature thresholds. Attributes such as '**SSLfinal_State**' and '**URL_Length**' appear near the root of the tree, confirming their strong predictive influence in early decision splits.

All three predictive models — **Logistic Regression**, **Decision Tree**, and **Random Forest** — proved to be reliable, as they all achieved high accuracy (92-97%) in classifying websites as either phishing or legitimate. Among these, **Random Forest** achieved the highest predictive performance. It balances precision and recalls very well for both classes, with few errors.

Models indicate that '**SSLfinal_State**' and '**URL_of_Anchor**' are top indicators for identifying phishing websites, as they were chosen by two of the top five indicators in all three models.

Conclusion

The analyses, descriptive, exploratory, and predictive analyses successfully demonstrated a clear, data-driven understanding of phishing website characteristics.

Descriptive Analysis showed that phishing websites typically have invalid SSL certificates, suspicious anchor links, long or abnormal URLs, and multiple subdomains. These features appeared frequently with phishing flags and formed the basis for identifying malicious sites.

Exploratory Analysis visually confirmed these findings from descriptive Analysis. Among all attributes, **SSLfinal_State** and **URL_of_Anchor** stood out as the strongest positive correlation with phishing behaviour, appearing as distinct visual patterns across multiple charts. This reinforces their reliability as consistent indicators of phishing.

Predictive Analysis validated these insights through three machine learning models. Logistic Regression and Decision Tree models achieved a strong accuracy of around 92%, while the **Random Forest** model achieved the highest performance (96.7%), balancing precision (0.97) and recall (0.95). The models support facts from exploratory Analysis that SSL and hyperlink attributes are the most influential in distinguishing phishing from legitimate sites.

From a cybersecurity perspective, these findings emphasise that automated website detection systems should prioritise features such as SSL validation and anchor links. Regular monitoring of these factors can enhance web-filtering systems and help organisations defend against phishing attacks.

In conclusion, integrating data analytics with machine learning can enhance cybersecurity threat detection, more effectively prevent phishing attacks, and reduce the risk of credential theft and financial fraud.

Resources and References

Dataset source:

Mohammad, R., & McCluskey, L. (2015, March 25). *Phishing Websites*. UC Irvine. Retrieved from https://archive.ics.uci.edu/dataset/327/phishing+websites

Tools:

Jupyter Notebook: https://jupyter.org/

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